# Limits Graphically and Numerically

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Estimating limits from a table**

A function $g$ satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \to 3} g(x)$.

Both sides approach $7$: $\lim_{x \to 3} g(x) = 7$.

**2** **[Foundational]** • **Target: Estimating limits from a table**

Using a table of values with $x$ near $0$ (radians), estimate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

Values approach $0.50$ from both sides, consistent with the exact limit $\frac{1}{2}$.

**3** **[Foundational]** • **Target: Reading limits from a graph**

The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ $(x \neq 1)$, state $\lim_{x \to 1} f(x)$.

$\lim_{x \to 1} f(x) = 2$ (even though $f(1)$ itself is undefined — a removable discontinuity).

**4** **[Foundational]** • **Target: Reading limits from a graph**

A function $h$ has $\lim_{x \to 2^-} h(x) = 5$ and $\lim_{x \to 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \to 2} h(x)$ and explain whether $h$ is continuous at $x = 2$.

$\lim_{x \to 2} h(x) = 5$, since both one-sided limits agree. But $h$ is NOT continuous at $x = 2$, since $\lim_{x \to 2} h(x) = 5 \neq h(2) = 8$ — this is a removable discontinuity.

**5** **[Foundational]** • **Target: One-sided limits and limits that fail to exist**

For $f(x) = \frac{x}{|x|}$, find:

**a** $\lim_{x \to 0^+} f(x) = 1$ **b** $\lim_{x \to 0^-} f(x) = -1$ **c** $\lim_{x \to 0} f(x)$ does not exist

(a) $= 1$; (b) $= -1$; (c) does not exist

**6** **[Foundational]** • **Target: One-sided limits and limits that fail to exist**

A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

$\lim_{x \to 1^-} f(x) = 1^2 = 1$. $\lim_{x \to 1^+} f(x) = 3 - 1 = 2$. Since the one-sided limits disagree $(1 \neq 2)$, $\lim_{x \to 1} f(x)$ does not exist.

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**7** [Foundational] • Target: Algebraic evaluation

Evaluate each limit by direct substitution or factoring:

**a** $\lim_{x\to 4}(3x^2-5x+1) \quad = 3(16)-20+1=29$

**b** $\lim_{x\to 3}\frac{x^2-9}{x-3} \quad = \lim_{x\to 3}(x+3)=6$

**c** $\lim_{x\to -2}\frac{x^2+5x+6}{x+2} \quad = \lim_{x\to -2}(x+3)=1$

(a) $=3(16)-20+1=29$; (b) $=\lim_{x\to 3}(x+3)=6$; (c) $=\lim_{x\to -2}(x+3)=1$

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Algebraic evaluation

Evaluate $\lim_{x\to 0}\frac{\sqrt{x+9}-3}{x}$ by rationalizing the numerator.

Multiply by $\frac{\sqrt{x+9}+3}{\sqrt{x+9}+3}$: $\frac{x}{x(\sqrt{x+9}+3)}=\frac{1}{\sqrt{x+9}+3}\to\frac{1}{6}$.

**9** [Intermediate] • Target: Limits at infinity from a graph

The graph shows $f(x)=\frac{2x}{x+1}$, which has a horizontal asymptote. Use the graph to estimate $\lim_{x\to\infty}f(x)$, then confirm algebraically by dividing numerator and denominator by $x$.

The graph flattens toward height $2$ as $x\to\infty$. Algebraically: $\frac{2x}{x+1}=\frac{2}{1+1/x}\to\frac{2}{1+0}=2$.

**10** [Intermediate] • Target: Limits involving infinite discontinuities

For $f(x)=\frac{1}{x-3}$, find $\lim_{x\to 3^-}f(x)$ and $\lim_{x\to 3^+}f(x)$, and state whether $\lim_{x\to 3}f(x)$ exists.

As $x\to 3^-$, $x-3\to 0^-$, so $f(x)\to -\infty$. As $x\to 3^+$, $x-3\to 0^+$, so $f(x)\to +\infty$. Since the one-sided limits are not equal (in fact neither is finite), $\lim_{x\to 3}f(x)$ does not exist.

**11** [Intermediate] • Target: Limits involving infinite discontinuities

Evaluate $\lim_{x\to\infty}\frac{3x^2+1}{x^2-4}$, using the fact that the numerator and denominator have the same degree.

For equal-degree rational functions, the limit is the ratio of leading coefficients: $\lim_{x\to\infty}\frac{3x^2+1}{x^2-4}=\frac{3}{1}=3$.

**12** [Intermediate] • Target: Estimating limits with a calculator-style table

Construct a mental table of values for $f(x)=\frac{e^x-1}{x}$ at $x=-0.1,\ -0.01, 0.01, 0.1$ (all close to $1$), and use it to estimate $\lim_{x\to 0}\frac{e^x-1}{x}$.

The values all cluster near $1$ (e.g. $f(0.1)\approx 1.052$, $f(-0.1)\approx 0.952$, both approaching $1$ as $x\to 0$): $\lim_{x\to 0}\frac{e^x-1}{x}=1$ (a standard limit related to the derivative of $e^x$ at $0$).

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**13** **[Intermediate]** • **Target: Limits with piecewise graphical behavior**

A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \le x < 3$, and $f(x) = 8 - x$ for $x \ge 3$. Find $\lim_{x \to 0} f(x)$ and $\lim_{x \to 3} f(x)$, stating whether each exists.

At $x = 0$: $\lim_{x \to 0^-} f(x) = 2$, $\lim_{x \to 0^+} f(x) = 0 + 2 = 2$: limit exists, equals $2$. At $x = 3$: $\lim_{x \to 3^-} f(x) = 3 + 2 = 5$, $\lim_{x \to 3^+} f(x) = 8 - 3 = 5$: limit exists, equals $5$.

**14** **[Intermediate]** • **Target: Limits with piecewise graphical behavior**

Evaluate $\lim_{x \to 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.

$\frac{(x-2)(x-3)}{(x-2)(x+2)} = \frac{x-3}{x+2} \to \frac{2-3}{2+2} = -\frac{1}{4}$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: One-sided limits and limits that fail to exist** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Consider the function $f(x)$ whose graph is shown below. The graph consists of a line segment and a quarter circle. The line segment connects the points $(2, 4)$ and $(4, 0)$. The quarter circle is in the second quadrant and is centered at $(0, 0)$ with radius 2. Let $A(x)$ be the area of the region bounded by the graph of $f(t)$, the $t$-axis, and the lines $t = 0$ and $t = x$. (a) Find $A(3)$.

The function $f(x)$ is defined piecewise. For $0 \le t \le 2$, $f(t)$ is the upper part of a circle centered at the origin with radius 2, so $f(t) = \sqrt{4 - t^2}$. For $2 < t \le 4$, $f(t)$ is a line segment connecting $(2, 4)$ and $(4, 0)$. The equation of this line can be found using the two-point form: $y - 0 = \frac{4-0}{2-4}(x - 4)$, which simplifies to $y = -2(x - 4)$, so $f(t) = -2(t - 4)$. To find $A(3)$, we need to calculate the definite integral $\int_0^3 f(t)\,dt$. We split this integral into two parts based on the definition of $f(t)$: $A(3) = \int_0^2 f(t)\,dt + \int_2^3 f(t)\,dt$ For the first part, $\int_0^2 \sqrt{4 - t^2}\,dt$, this represents the area of a quarter circle of radius 2, which is $\frac{1}{4}\pi r^2 = \frac{1}{4}\pi(2^2) = \pi$. For the second part, $\int_2^3 (-2(t - 4))\,dt$, we can evaluate this integral: $\int_2^3 (-2t + 8)\,dt = [-t^2 + 8t]_2^3$ Evaluating at the limits: $(-3^2 + 8 \cdot 3) - (-2^2 + 8 \cdot 2)$ $(-9 + 24) - (-4 + 16)$ $15 - 12 = 3$. Therefore, $A(3) = \pi + 3$. *Common Misconceptions:* Students might incorrectly assume the function is a single simple shape or miscalculate the line segment's equation. They might also forget that the first part of the integral represents a quarter circle's area and attempt to integrate $\sqrt{4 - t^2}$ directly, which is more complex.

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**16** **[Challenging]** • **Target: Algebraic evaluation** • **Cross-Lesson: 2.01 The Definition of the Derivative**

The graph of the derivative of a function $f(x)$, denoted by $f'(x)$, is shown in the figure above. The function $f(x)$ is continuous, and it is known that $f(1) = 5$. Determine the value of $f(5)$.

The Fundamental Theorem of Calculus states that $\int_a^b f'(x)\,dx = f(b) - f(a)$. Rearranging this, we get $f(b) = f(a) + \int_a^b f'(x)\,dx$. In this problem, we are given $f(1) = 5$ and we need to find $f(5)$. Thus, we have $a = 1$ and $b = 5$. We need to calculate the definite integral $\int_1^5 f'(x)\,dx$, which represents the area under the curve of $f'(x)$ from $x = 1$ to $x = 5$. From the graph, $f'(x)$ is a straight line. The area from $x = 1$ to $x = 5$ can be split into two parts: a triangle and a rectangle. Part 1: Area from $x = 1$ to $x = 3$. This is a triangle with base $3 - 1 = 2$ and height $f'(1) = 3$. The area of this triangle is $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2 \times 3 = 3$. Part 2: Area from $x = 3$ to $x = 5$. This is a rectangle with base $5 - 3 = 2$ and height $f'(3) = 3$. The area of this rectangle is $\text{base} \times \text{height} = 2 \times 3 = 6$. Therefore, the total area under the curve from $x = 1$ to $x = 5$ is $\int_1^5 f'(x)\,dx = 3 + 6 = 9$. Now, we can find $f(5)$: $f(5) = f(1) + \int_1^5 f'(x)\,dx = 5 + 9 = 14$. Common traps: Students might misinterpret the graph and assume $f'(x)$ is constant or linear over the entire interval without splitting it. They might also make calculation errors with the area of the triangle or rectangle.

**17** **[Challenging]** • **Target: Limits at infinity from a graph** • **Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives**

The graph of the derivative of a function $f(x)$ is shown below. Given that $f(2) = 5$, determine the value of $f(6)$.

The graph shows $f'(x)$. By the Fundamental Theorem of Calculus, $f(6) - f(2) = \int_2^6 f'(x)\,dx$. The integral $\int_2^6 f'(x)\,dx$ represents the area under the curve of $f'(x)$ from $x = 2$ to $x = 6$. This area is a trapezoid with parallel sides of length $4$ (from $x = 2$ to $x = 4$) and $6$ (from $x = 4$ to $x = 6$), and a height of $2$ (the difference in x-values from 2 to 4 and 4 to 6, but looking at the graph the height from y=1 to y=5 is 4 and from y=1 to y=6 is 5, the vertical distance represents the change in y for the integral, which is the change in x on the graph. The area is a trapezoid with heights of 4 and 5 at x=2 and x=6 respectively, and a base of 4. The area under the curve from x=2 to x=6 is the area of a trapezoid with parallel sides of length 4 (at x=2, f'(2)=4) and 5 (at x=6, f'(6)=5), and a height (width along x-axis) of 6-2=4. The area is $\frac{1}{2}(4 + 5) \times 4 = \frac{1}{2}(9) \times 4 = 18$. Therefore, $f(6) - f(2) = 18$. Given $f(2) = 5$, we have $f(6) - 5 = 18$, which means $f(6) = 18 + 5 = 23$. Common algebraic trap: Students might misinterpret the y-axis values as the length of the parallel sides of the trapezoid and the x-axis values as the height, or vice versa. They might also forget to add the initial value of $f(2)$.

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**18** **[Challenging]** • **Target: Limits involving infinite discontinuities** • **Cross-Lesson: 3.01 The Chain Rule**

Consider a function $f(x)$ whose derivative is given by $f'(x) = 6x^2 - 12x + 6$. The graph of the function $g(x) = x^3 - 6x^2 + 9x - 2$ is shown below. If $f(x)$ and $g(x)$ have the same tangent line at $x = 1$, determine the value of $f''(0)$.

First, let's find the derivative of $g(x)$: $g'(x) = 3x^2 - 12x + 9$. At $x = 1$, the slope of the tangent line to $g(x)$ is $g'(1) = 3(1)^2 - 12(1) + 9 = 3 - 12 + 9 = 0$. The problem states that $f(x)$ and $g(x)$ have the same tangent line at $x = 1$. This means $f'(1) = g'(1) = 0$. We are given $f'(x) = 6x^2 - 12x + 6$. Let's check: $f'(1) = 6(1)^2 - 12(1) + 6 = 6 - 12 + 6 = 0$. This confirms consistency. To find the value of $f''(0)$, we first need to find the second derivative of $f(x)$. We are given $f'(x) = 6x^2 - 12x + 6$. Therefore, $f''(x) = \frac{d}{dx}(6x^2 - 12x + 6) = 12x - 12$. Now, we can evaluate $f''(0)$: $f''(0) = 12(0) - 12 = -12$. Common misconception: Students might try to integrate $f'(x)$ to find $f(x)$ and then differentiate, or they might confuse the derivative of $f(x)$ with the derivative of $g(x)$ when calculating the second derivative. The value of $g(x)$ at $x = 1$ is $g(1) = 1^3 - 6(1)^2 + 9(1) - 2 = 1 - 6 + 9 - 2 = 2$. This information is not directly needed to find $f''(0)$ given the provided information about the tangent line slope.

**19** **[Challenging]** • **Target: Estimating limits with a calculator-style table** • **Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Consider a function $f(x)$ such that its derivative is given by $f'(x) = 3x^2 - 12x + 10$. The graph of the second derivative, $f''(x)$, is shown below. Given that $f(0) = 5$, find the value of $f(2)$.

The derivative of $f'(x)$ is $f''(x)$. From the graph, we can observe that $f''(x)$ is a parabola opening upwards with its vertex at $(2, -2)$. The equation of a parabola with vertex $(h, k)$ is $a(x - h)^2 + k$. Thus, $f''(x) = a(x - 2)^2 - 2$. We can also see that the parabola passes through $(0, 10)$, so $10 = a(0 - 2)^2 - 2$ , which gives $12 = 4a$, so $a = 3$. Therefore, $f''(x) = 3(x - 2)^2 - 2 = 3(x^2 - 4x + 4) - 2 = 3x^2 - 12x + 12 - 2 = 3x^2 - 12x + 10$. This matches the given $f'(x)$, confirming our observation of $f''(x)$. Now we need to find $f(x)$ from $f'(x)$. We integrate $f'(x)$ with respect to $x$: $f(x) = \int (3x^2 - 12x + 10)\,dx$ $f(x) = x^3 - 6x^2 + 10x + C$ We are given that $f(0) = 5$. Substituting $x = 0$ into the equation for $f(x)$: $f(0) = (0)^3 - 6(0)^2 + 10(0) + C$ $5 = C$ So, the function is $f(x) = x^3 - 6x^2 + 10x + 5$. Finally, we need to find $f(2)$: $f(2) = (2)^3 - 6(2)^2 + 10(2) + 5$ $f(2) = 8 - 6(4) + 20 + 5$ $f(2) = 8 - 24 + 20 + 5$ $f(2) = -16 + 25$ $f(2) = 9$.

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**20** [Challenging] • Target: Limits with piecewise graphical behavior • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

The graph of a differentiable function $f$ is shown in the figure above. The function $f$ has a local maximum at $x = c$ and a local minimum at $x = e$. The point $(c, f(c))$ is approximately $(1, 4)$ and the point $(e, f(e))$ is approximately $(4, -2)$. The shaded region, bounded by the graph of $f$, the x-axis, and the lines $x = 0$ and $x = 5$, has area 12. If $\int_0^c f(x)\,dx = k$, what is the value of $\int_c^5 f(x)\,dx$ in terms of $k$?

The total area of the shaded region from $x = 0$ to $x = 5$ is given as 12. This total area can be expressed as the sum of the areas of two sub-regions: the area from $x = 0$ to $x = c$ and the area from $x = c$ to $x = 5$. Mathematically, this can be written using definite integrals: $\int_0^5 f(x)\,dx = \int_0^c f(x)\,dx + \int_c^5 f(x)\,dx$. We are given that the total area is 12, so $\int_0^5 f(x)\,dx = 12$. We are also given that $\int_0^c f(x)\,dx = k$. Substituting these values into the equation: $12 = k + \int_c^5 f(x)\,dx$. To find the value of $\int_c^5 f(x)\,dx$, we can rearrange the equation: $\int_c^5 f(x)\,dx = 12 - k$. Therefore, the value of $\int_c^5 f(x)\,dx$ in terms of $k$ is $12 - k$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Estimating limits from a table

Find $\lim_{x \to -3} (x^2 + 3x - 4)$.

**a.** $-3$

**b.** $-4$ ✓

**c.** $4$

**d.** $-5$

Direct substitution: $1(-3)^2 + 3(-3) - 4 = -4$.

**22** [Foundational] • Target: Reading limits from a graph

Find $\lim_{x \to 3} (4x^2 - x + 8)$.

**a.** $41$ ✓

**b.** $42$

**c.** $40$

**d.** $-41$

Direct substitution: $4(3)^2 - (3) + 8 = 41$.

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**23** [Foundational] • Target: One-sided limits and limits that fail to exist

Find $\lim_{x \to 3} (5x^2 + 7)$.

**a.** $-52$

**b.** $51$

**c.** $52$ ✓

**d.** $53$

Direct substitution: $5(3)^2 + 7 = 52$.

**24** [Foundational] • Target: Algebraic evaluation

Find $\lim_{x \to 4} (5x^2 - 8x + 5)$.

**a.** $54$

**b.** $53$ ✓

**c.** $-53$

**d.** $52$

Direct substitution: $5(4)^2 - 8(4) + 5 = 53$.

**25** [Foundational] • Target: Limits at infinity from a graph

Find $\lim_{x \to -2} (x^2 - 4x + 6)$.

**a.** $17$

**b.** $19$

**c.** $-18$

**d.** $18$ ✓

Direct substitution: $1(-2)^2 - 4(-2) + 6 = 18$.

**26** [Foundational] • Target: Limits involving infinite discontinuities

Find $\lim_{x \to 3} (4x^2 + 3x - 5)$.

**a.** $39$

**b.** $40$ ✓

**c.** $41$

**d.** $-40$

Direct substitution: $4(3)^2 + 3(3) - 5 = 40$.

**27** [Foundational] • Target: Estimating limits with a calculator-style table

Find $\lim_{x \to 1} (2x^2 - x - 3)$.

**a.** $2$

**b.** $-3$

**c.** $-2$ ✓

**d.** $-1$

Direct substitution: $2(1)^2 - (1) - 3 = -2$.

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**28** [Foundational] • Target: Limits with piecewise graphical behavior

Evaluate $\lim_{x\to 0} \frac{(x)(x-1)}{x}$ by factoring and cancelling.

**a.** $-3$

**b.** $1$

**c.** $-1$ ✓

**d.** $3$

Cancel the common factor: $\frac{(x)(x-1)}{x} = x - 1$ for $x \neq 0$, so the limit is $0 - (1) = -1$.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Estimating Limits Numerically & Tables

Evaluate $\lim_{x\to 5} \frac{(x-5)(x-6)}{x-5}$ by factoring and cancelling.

**a.** $-1$ ✓

**b.** $1$

**c.** $11$

**d.** $-3$

Cancel the common factor: $\frac{(x-5)(x-6)}{x-5} = x - 6$ for $x \neq 5$, so the limit is $5 - (6) = -1$.

**30** [Intermediate] • Target: Graphical Analysis & Removable Discontinuities

Evaluate $\lim_{x\to -4} \frac{(x+4)(x-5)}{x+4}$ by factoring and cancelling.

**a.** $-11$

**b.** $-7$

**c.** $-9$ ✓

**d.** $9$

Cancel the common factor: $\frac{(x+4)(x-5)}{x+4} = x - 5$ for $x \neq -4$, so the limit is $-4 - (5) = -9$.

**31** [Intermediate] • Target: One-Sided Limits & Absolute Value Quotients

Evaluate $\lim_{x\to -1} \frac{(x+1)(x-6)}{x+1}$ by factoring and cancelling.

**a.** $-9$

**b.** $-5$

**c.** $7$

**d.** $-7$ ✓

Cancel the common factor: $\frac{(x+1)(x-6)}{x+1} = x - 6$ for $x \neq -1$, so the limit is $-1 - (6) = -7$.

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**32** [Intermediate] • Target: Infinite Limits & Horizontal Asymptotes

Evaluate $\lim_{x\to 6} \frac{(x-6)(x-3)}{x-6}$ by factoring and cancelling.

**a.** $1$

**b.** $-3$

**c.** $3$ ✓

**d.** $5$

Cancel the common factor: $\frac{(x-6)(x-3)}{x-6} = x-3$ for $x \neq 6$, so the limit is $6-(3)=3$.

**33** [Intermediate] • Target: Estimating limits from a table

Evaluate $\lim_{x\to 3} \frac{(x-3)(x+3)}{x-3}$ by factoring and cancelling.

**a.** $4$

**b.** $-6$

**c.** $8$

**d.** $6$ ✓

Cancel the common factor: $\frac{(x-3)(x+3)}{x-3} = x+3$ for $x \neq 3$, so the limit is $3-(-3)=6$.

**34** [Intermediate] • Target: Reading limits from a graph

Evaluate $\lim_{x\to -5} \frac{(x+5)(x+1)}{x+5}$ by factoring and cancelling.

**a.** $-4$ ✓

**b.** $-6$

**c.** $-2$

**d.** $4$

Cancel the common factor: $\frac{(x+5)(x+1)}{x+5} = x+1$ for $x \neq -5$, so the limit is $-5-(-1)=-4$.

**35** [Intermediate] • Target: One-sided limits and limits that fail to exist

Evaluate $\lim_{x\to \infty} \frac{6x^2+3x}{8x^2+8x}$.

**a.** $\frac{4}{3}$

**b.** $\frac{3}{4}$ ✓

**c.** $\frac{7}{8}$

**d.** $\frac{2}{3}$

Numerator and denominator have equal degree $(2)$, so the limit is the ratio of leading coefficients: $\frac{6}{8}=\frac{3}{4}$.

**36** [Intermediate] • Target: Algebraic evaluation

Evaluate $\lim_{x\to \infty} \frac{5x^2+6x}{9x^2-3x}$.

**a.** $\frac{5}{9}$ ✓

**b.** $\frac{1}{2}$

**c.** $\frac{9}{5}$

**d.** $\frac{2}{3}$

Numerator and denominator have equal degree $(2)$, so the limit is the ratio of leading coefficients: $\frac{5}{9}=\frac{5}{9}$.

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**37** [Intermediate] • Target: Limits at infinity from a graph

Evaluate $\lim_{x\to\infty} \frac{6x+3}{7x^2-1}$.

**a.** does not exist

**b.** 0 ✓

**c.** $\frac{6}{7}$

**d.** 1

The denominator's degree (2) exceeds the numerator's (1), so the quotient shrinks to 0 as $x \to \infty$.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Limits involving infinite discontinuities • Cross-Lesson: 3.01 The Chain Rule

For $f(x) = \frac{-1}{x-2}$, find $\lim_{x\to 2^-} f(x)$.

**a.** 2

**b.** 0

**c.** $-\infty$

**d.** $+\infty$ ✓

As $x \to 2^-$, the denominator $\to 0^-$, so $f(x) \to +\infty$.

**39** [Challenging] • Target: Estimating limits with a calculator-style table • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Evaluate $\lim_{x\to\infty} \frac{8x+3}{x^2-1}$.

**a.** does not exist

**b.** 8

**c.** 0 ✓

**d.** 1

The denominator's degree (2) exceeds the numerator's (1), so the quotient shrinks to 0 as $x \to \infty$.

**40** [Challenging] • Target: Limits with piecewise graphical behavior • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Evaluate $\lim_{x\to\infty} \frac{8x^3+2x}{2x^2+1}$.

**a.** 0

**b.** 1

**c.** **does not exist ($+\infty$)** ✓

**d.** 4

The numerator's degree (3) exceeds the denominator's (2), so the quotient grows without bound: the limit does not exist (diverges to $+\infty$).

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**41** [Challenging] • Target: Estimating Limits Numerically & Tables • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

A function $g$ satisfies $g(1.9) = 3.81, g(1.99) = 3.98, g(2.01) = 4.02,$ and $g(2.1) = 4.19$. What is the estimated value of $\lim_{x \to 2} g(x)$?

**a.** 3.9

**b.** 4.0 ✓

**c.** 4.1

**d.** Does not exist

As $x$ approaches $2$ from both the left $(1.9, 1.99)$ and right $(2.1, 2.01)$, the values of $g(x)$ consistently approach $4.0$.

**42** [Challenging] • Target: Estimating Limits Numerically & Tables • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Values of a function $h(x)$ near $x = 3$ are $h(2.99) = 5.01$ and $h(3.01) = -4.99$. What does this suggest about $\lim_{x \to 3} h(x)$?

**a.** The limit is $0$

**b.** **The limit does not exist because left and right limits disagree** ✓

**c.** The limit is $5$

**d.** The function is continuous at $x = 3$

Because the left-hand limit approaches $+5$ while the right-hand limit approaches $-5$, the two-sided limit does not exist.

**43** [Challenging] • Target: Graphical Analysis & Removable Discontinuities • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Which of the following functions has a vertical asymptote at $x = 2$ and NOT a removable hole?

**a.** $f(x) = \frac{x-2}{x^2-4}$

**b.** $f(x) = \frac{x+1}{(x-2)^2}$ ✓

**c.** $f(x) = \frac{x^2-4}{x-2}$

**d.** $f(x) = \frac{2x-4}{x-2}$

In $f(x) = \frac{x+1}{(x-2)^2}$, the factor $(x-2)$ does not cancel with the numerator, creating a vertical asymptote at $x = 2$.

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**44** [Challenging] • Target: One-Sided Limits & Absolute Value Quotients • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Evaluate: $\lim_{x \to 5^-} \frac{x-5}{|x-5|}$.

**a.** 1

**b.** $-1$ ✓

**c.** 0

**d.** Does not exist

For $x < 5$, $x - 5 < 0$, which means $|x-5| = -(x-5)$. Thus $\frac{x-5}{-(x-5)} = -1$.

**45** [Challenging] • Target: Infinite Limits & Horizontal Asymptotes • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Evaluate: $\lim_{x \to 3^+} \frac{1}{x-3}$.

**a.** $\infty$ ✓

**b.** $-\infty$

**c.** 0

**d.** 1

As $x \to 3$ from the right, the denominator is positive and approaches $0$, so the quotient grows without bound toward $\infty$.

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